Managing the AI Transition: Overshoot, Non-Replacement, and Adaptive Policy

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Author note. This manuscript develops an original policy framework for AI-driven labor-market transition. It was revised to add citable research objects, a formal transition-overshoot metric, calibrated simulations, a completed public-data JOLTS exercise, and a referee-style senior-economist review. The empirical results are reported transparently: the public JOLTS data show suggestive post-2022 declines in hires per opening in AI-exposed sectors but do not yet deliver robust causal proof at broad industry aggregation.

Abstract

This paper argues that AI-driven labor disruption should be modeled not only as a static substitution or automation-tax problem, but as a dynamic transition-control problem. Competitive AI adoption may generate transition overshoot: labor demand can fall faster than demand, human capital, firm capacity, and policy institutions can adjust. The paper introduces a dampening framework, a Transition Overshoot Index, the non-replacement mechanism, and the capability-deployment gap as reusable research objects. A calibrated simulation compares laissez-faire, static Pigouvian-style intervention, and adaptive dampening. A completed public-data JOLTS exercise estimates hires/openings difference-in-differences specifications using BLS monthly industry data through March 2026. The JOLTS results are suggestive but not causally decisive: broad AI-exposed sectors exhibit lower post-2022 hires/openings ratios, but the two-way fixed-effect DiD coefficient is not statistically significant. The contribution is therefore best read as a foundational framework and research agenda disciplined by first-stage public evidence, not as final causal proof. The final revision adds a clean feedback-dominance proposition, a stronger firm-level evidence synthesis using payroll, job-posting, resume, Compustat, and Revelio-style evidence from the literature, and an explicit data-access protocol for converting the framework into a fully causal firm-level design.

A further extension added in this version is the capital-market growth-pressure cascade: AI-linked workforce compression can diffuse from high-growth frontier technology firms to peer and lower-tier firms through investor expectations, managerial imitation, and AI-efficiency signaling. This mechanism helps explain why the negative trough in labor demand can form before firms learn that they have compressed beyond the efficient transition path.

Keywords: artificial intelligence; labor displacement; automation; coordination failure; optimal taxation; control theory; unemployment; JOLTS; economic transition

JEL classifications: E24, E61, H23, J23, J24, O33

1. Introduction

The economic question raised by artificial intelligence is often framed too narrowly. One side asks whether AI will eliminate jobs; the other answers that technology has always created new work. Both frames miss the transition itself. Even if a new long-run equilibrium eventually contains substantial human employment, the path to that equilibrium can involve concentrated unemployment, reduced labor-market entry for young

workers, depressed demand, firm failures, and political instability. These are not side issues. They are the welfare-relevant objects of policy.

This paper argues that the current AI transition should be analyzed as a problem of dynamic coordination under uncertainty. Firms adopting AI face private incentives that differ from the collective interest of the sector. A firm that automates captures the full cost saving from substituting AI for labor, but the demand loss created when workers lose income is spread across the market. This is the over-automation logic developed by Falk and Tsoukalas (2026). The present paper accepts that result and extends it from a static equilibrium comparison to a transition-management problem.

The practical concern is not only that firms may automate too much in the long run. It is that they may automate too quickly, through mechanisms that standard data understate, while policy instruments respond too slowly. A transition that is efficient in a frictionless model can be destructive when workers cannot retrain instantly, firms cannot costlessly survive demand collapses, and governments observe the damage with lags. This is the core motivation for the dampening framework.

The term 'dampening' is used deliberately. The aim is not to abolish AI, to freeze the economy at the pre-AI allocation, or to prevent productivity gains. The aim is to reduce destructive oscillation, peak harm, and irreversible damage during the transition. A well-designed dampening regime slows the privately excessive margin of automation, supports purchasing power while workers adjust, accelerates genuine reallocation, and preserves productive capacity. It works like a shock absorber rather than a prohibition.

The paper makes five contributions. First, it identifies the non-replacement mechanism: firms may reduce labor demand not primarily by firing workers, but by failing to backfill positions when workers leave and by hiring AI-deployment specialists who enable further headcount compression. This mechanism can be consistent with stable aggregate employment and concentrated deterioration in entry-level opportunities. Second, it formalizes the capability-deployment gap: the economically relevant AI parameter is not benchmark capability alone, but task-specific reliable deployability. Third, it recasts AI labor policy as an optimal control problem focused on transient response. Fourth, it proposes a feasible policy portfolio, including an administratively practical revenue-based automation tax. Fifth, it provides a replicable empirical design using public data to test the non-replacement hypothesis.

The paper is intentionally positioned as a working paper. Its formal model is minimal rather than exhaustive, and its empirical section is a design and implementation blueprint rather than a claim of causal proof. That is the appropriate posture for a phenomenon still unfolding. The objective is to make the argument rigorous enough to be debated, tested, and improved.

Contribution	What it adds	Testable implication
Non-replacement mechanism	Labor demand falls through attrition non-backfill and altered hiring, not only layoffs	AI-exposed sectors show declining hires per opening and weaker entry-level hiring
Capability-deployment gap	Effective AI substitutability depends on reliable deployment, not benchmarks	Adoption and displacement vary by task verifiability and error cost
Dampening framework	Policy is a control problem over the transition path	Adaptive combined instruments produce shallower employment troughs
Revenue-based automation tax	Administrative feasibility is treated as part of welfare design	Coarser tax bases may outperform precise but unenforceable task taxes

1.1 Central contribution and citable research objects

The paper is organized around one central claim: AI labor disruption is not merely an equilibrium substitution problem; it is a dynamic transition-management problem. The foundational object is the transition path, not

only the endpoint. Competitive AI adoption may move the economy toward a more productive long-run allocation while still generating excessive short-run labor-demand loss, demand collapse, firm exit, and human-capital scarring along the way.

The paper therefore introduces a research program that other scholars can cite, test, and extend. The terms below are defined to be portable rather than rhetorical. Each can be used in theoretical models, empirical designs, or policy evaluation.

Concept	Definition for future citation	How researchers can build on it
Transition overshoot problem	The welfare-relevant AI-labor problem in which the adjustment path temporarily moves labor demand, demand, or productive capacity beyond the socially efficient transition path, even if the long-run AI equilibrium is desirable.	Estimate overshoot across sectors; compare countries or policies; integrate with macro models of hysteresis and adjustment costs.
Transition Overshoot Index (TOI)	A normalized measure of the maximum realized transition shortfall relative to the expected long-run adjustment.	Use as an outcome variable in event studies, cross-country comparisons, and policy evaluations.
Non-replacement mechanism	A labor-demand reduction channel in which firms shrink effective headcount through attrition non-backfill and AI-complement hiring rather than visible layoffs.	Measure with hires/openings, hires/separations, entry-level hiring, and revenue-per-employee patterns.
Capability-deployment gap	The wedge between benchmark AI capability and reliable production deployment after verification, accountability, integration, and error-cost frictions.	Estimate by task type; use to explain heterogeneous adoption and displacement speeds.
Dampening framework	A policy framework that treats AI labor policy as control of the transition path rather than only correction of the static externality.	Derive optimal controls, estimate policy rules, and test adaptive regimes against static policies.
Adaptive intervention	A rule in which policy intensity responds to observed transition stress and relaxes as the system stabilizes.	Compare adaptive taxes, wage insurance, UBI triggers, and retraining triggers using simulations and natural experiments.
Revenue-based automation tax	An administratively feasible tax base tied to AI-attributable revenue rather than unobservable per-task automation.	Study incidence, avoidance, international coordination, and second-best welfare performance.
JOLTS hires/openings test	A first-pass public-data test for non-replacement using divergence in hires-per-opening ratios between AI-exposed and comparison sectors.	Implement difference-in-differences, event studies, placebo tests, and alternative AI-exposure measures.
Calibrated dampening simulation	A tractable numerical environment comparing laissez-faire, static dampening, and adaptive dampening under empirically interpretable parameters.	Replicate, structurally estimate, extend to sectors, add expectations, and test robustness.

The following concepts are stated in portable form so that other researchers can cite, measure, criticize, or extend them.

Over-automation coordination failure. The paper treats AI adoption as a strategic interaction in which each firm captures private labor-cost savings while externalizing part of the demand loss created by displaced or non-hired workers. The important extension here is dynamic: the coordination failure does not merely select

the wrong equilibrium; it can also cause firms collectively to discover the new equilibrium through excessive short-run labor-demand destruction.

Non-replacement mechanism. The paper defines non-replacement as labor-demand reduction that occurs when separations, retirements, or anticipated positions are not backfilled because AI tools absorb the marginal work. This mechanism is empirically important because it can reduce opportunity without generating layoffs, WARN notices, or immediate unemployment spikes. It is proposed as a bridge between aggregate-stable labor data and cohort-specific entry-level stress.

Capability-deployment gap. The paper distinguishes benchmark capability from production reliability. Economically relevant AI productivity is not what a model can do in a benchmark, but what it can do reliably in an audited production workflow after verification, liability, security, and human oversight costs. This gap explains why labor displacement can be gradual and sectorally heterogeneous even when headline AI capability rises quickly.

Dampening framework. The central policy framework treats AI transition management as a control problem. The objective is not to stop automation but to reduce peak labor-demand shortfall, avoid hysteresis, preserve firm capacity, and allow beneficial productivity gains to arrive without a destructive discovery process.

Adaptive intervention. The paper defines adaptive intervention as a state-contingent policy rule that responds to measured displacement velocity, hires/openings deterioration, income-replacement gaps, and sector-specific exposure. Adaptive policy is contrasted with fixed taxes or purely ex post redistribution.

Revenue-based automation tax. The paper proposes a revenue-based tax on AI-attributable output as an administratively feasible second-best instrument. It is not claimed to be theoretically first-best. Its purpose is to create a measurable brake and funding base when per-task Pigouvian taxation is too difficult to observe or audit.

Empirical JOLTS test. The paper operationalizes non-replacement through hires per job opening, using BLS JOLTS industry-level hires and openings. A persistent fall in hires per opening in AI-exposed sectors, especially when output per worker rises, is treated as suggestive evidence of non-replacement, subject to confounding and identification limits.

Simulation model. The simulation model is a calibrated laboratory for comparing policy regimes, not a structural estimate of the U.S. economy. Its role is to show how the same externality can produce different transition losses under laissez-faire, static intervention, and adaptive dampening, and to make those differences measurable through the Transition Overshoot Index.

1.2 Main propositions

Proposition	Statement	Status in this paper
P1: Dynamic overshoot	Competitive AI adoption can produce transition overshoot when firms internalize only part of the demand loss and labor-market recovery is delayed.	Derived in the minimal model and illustrated in simulations.
P2: Static correction is incomplete	A static Pigouvian tax can correct the contemporaneous automation externality but need not minimize transition welfare loss when adjustment costs, feedback delays, or irreversibility are present.	Shown analytically as a second-best transition problem.
P3: Adaptive dampening can dominate fixed policy	Under convex transition harm and bounded parameter uncertainty, an adaptive policy rule can reduce peak	Supported by calibrated simulations and robustness checks; not claimed as global optimality.

	shortfall relative to laissez-faire and static policy in the calibrated parameter region.	
P4: Non-replacement makes displacement under-observed	When labor demand declines through attrition non-backfill, standard layoff and unemployment measures understate effective displacement.	Formalized and linked to JOLTS proxies.
P5: Deployment reliability governs timing	Observed displacement depends on production reliability, not benchmark capability alone, so verification costs and accountability frictions produce sectoral heterogeneity.	Formalized through the capability-deployment parameter.

1.3 Static Pigouvian policy versus dynamic dampening

Dimension	Static externality / Pigouvian view	Dynamic dampening view
Main distortion	Private automation benefit exceeds privately internalized demand loss.	Externality plus overshoot, delay, hysteresis, and transition irreversibility.
Policy target	Correct the marginal private cost of automation.	Shape the path of adjustment while preserving long-run productivity.
Representative instrument	Per-task Pigouvian automation tax.	Adaptive portfolio: tax/brake, income support, retraining, coordination, financial-stability safeguards.
Welfare concern	Wrong equilibrium allocation.	Destructive transition path and threshold harms.
Failure mode	Over-automation in steady state.	Peak unemployment, demand collapse, firm exit, human-capital scarring, policy-delay instability.

2. What the Literature Has Established

2.1 The task-based foundation

The conceptual foundation for modern analysis of automation is the task-based framework developed by Acemoglu and Restrepo (2018, 2019), building on Zeira (1998), Autor, Levy, and Murnane (2003), and Acemoglu and Autor (2011). Production is decomposed into tasks. Automation shifts tasks from labor to capital, while new-task creation reinstates labor demand. Employment, wages, and labor share depend on the balance between displacement and reinstatement.

This task-level framing matters for AI. AI rarely replaces a whole occupation in one step. It substitutes for some tasks inside an occupation, complements others, and changes the minimum team size required to produce a given output. The policy problem therefore concerns labor demand at the task, role, team, and sector levels rather than only headline unemployment.

2.2 Competitive over-automation

Falk and Tsoukalas (2026) provide the most direct theoretical anchor for this paper. In their model, workers are also customers. A firm that replaces labor with AI captures the cost saving, but the demand loss caused by displaced workers' lost income is spread across all firms. The private incentive to automate therefore exceeds the collective incentive. The model implies a prisoner's-dilemma structure: individual firms have a dominant strategy to automate, while the sector can be worse off when all do so.

The model has two implications that are especially important here. First, competition can worsen over-automation because each firm internalizes a smaller fraction of the demand loss. Second, better AI can amplify

the problem by increasing the private cost saving and accelerating displacement. The authors evaluate several policy responses and find that only a Pigouvian automation tax directly closes the static wedge. This paper accepts the static result and asks a different question: how should policy manage the transition path when adjustment, measurement, political constraints, and deployment reliability are imperfect?

2.3 Empirical evidence on AI labor effects

The empirical evidence through early 2026 is mixed in a way that is informative rather than contradictory. Controlled deployment studies show that generative AI can materially raise productivity in particular tasks. Brynjolfsson, Li, and Raymond (2025), for example, document productivity gains for customer-support agents, with larger gains among less-experienced workers. Studies of software development and consulting similarly show meaningful task-level productivity effects, though the size and direction of effects vary across task types.

At the aggregate labor-market level, the evidence remains more muted. Humlum and Vestergaard (2025) link AI chatbot adoption surveys to Danish administrative labor records and estimate precise null or small effects on earnings and hours in the early period. At the same time, Brynjolfsson, Chandar, and Chen (2025) find concentrated declines for early-career workers in highly AI-exposed occupations. This coexistence of aggregate stability and cohort-specific stress is exactly the pattern the non-replacement mechanism would predict.

2.4 Optimal automation taxation

Guerreiro, Rebelo, and Teles (2022) show that robot taxation can be optimal when automation harms workers who cannot quickly adjust their skills. Their result is dynamic: robot taxes are highest while affected cohorts remain in the labor force and decline as those cohorts exit. This supports one of the central claims of the present paper: the optimal policy is unlikely to be a fixed permanent rate. Time-varying intervention is not a political compromise; it follows naturally from dynamic adjustment frictions.

Taken together, the literature establishes a strong foundation: AI should be analyzed at the task level; competitive automation can produce a coordination failure; early empirical effects are real but uneven; and optimal taxation can be time-varying. The present paper extends this foundation by focusing on mechanisms and transient dynamics that are not central in the existing models.

3. Three Gaps in Existing Analysis

3.1 The non-replacement mechanism

Most models treat displacement as an explicit employment reduction. Firms decide to fire or retain workers. That is analytically clean, but it may not describe the dominant early mechanism in AI-exposed knowledge work. Many firms can reduce labor demand by allowing natural attrition to shrink teams, by not backfilling departures, by slowing entry-level hiring, or by replacing a departed generalist with an AI-integration specialist who enables a smaller team to maintain output.

This matters because non-replacement is much less visible than layoffs. It does not necessarily generate WARN notices, restructuring announcements, or unemployment spells attributable to AI. A worker who would have been hired but is not hired does not appear in layoff data. A team that maintains output with eight workers instead of ten after two departures has reduced labor demand, but standard data may record only ordinary separations and ordinary vacancies.

The mechanism reconciles otherwise puzzling evidence. Aggregate employment can remain stable while entry-level opportunities deteriorate. Older workers can remain employed because incumbents are not being fired, while younger workers face weaker hiring pipelines. Standard unemployment statistics can therefore understate the true reduction in labor demand.

3.2 The capability-deployment gap

Formal models often treat AI capability as a single productivity parameter. In practice, the economically relevant parameter is effective reliable deployability. A system may perform well on benchmarks and still require costly human oversight in production, especially where errors are hard to detect, legally consequential, or operationally irreversible. Conversely, for verifiable and low-error-cost tasks, the gap between capability and deployment may be small.

This gap changes transition dynamics. Benchmark capability determines what appears possible; deployment reliability determines what firms actually substitute for labor. The gap therefore slows displacement in high-stakes tasks, accelerates it in verifiable tasks, and makes adoption heterogeneous across sectors. It also explains why human-in-the-loop roles may persist longer than headline capability forecasts suggest.

3.3 The time-dynamics problem

Equilibrium comparisons are necessary but incomplete. Welfare losses from technological transitions often occur during the path between equilibria: unemployment spells, lost labor-market entry, bankruptcies, demand collapses, financial stress, and political backlash. A policy that is optimal in steady state can be badly timed for the transition.

AI makes this timing problem acute because adoption can move faster than labor-market institutions. Firms can deploy tools within months. Retraining takes years. Tax law changes slowly. Public data arrive with lags. This mismatch creates a control problem: policy must act under uncertainty with delayed measurement and instruments of different speeds.

4. A Minimal Formal Model of Dampened Transition

4.1 Static externality and the over-automation wedge

Consider an industry with N symmetric firms. Each firm can automate a measure a_i of tasks previously performed by labor. Let s denote the private cost saving per automated task, and let D(a) denote the aggregate demand loss per automated task caused by reduced labor income. The firm internalizes only 1/N of the demand loss.

$$\text{Private automation condition:} \quad s > D(a)/N \qquad (1)$$

$$\text{Social automation condition:} \quad s > D(a) \qquad (2)$$

The gap between the social and private conditions is the uninternalized demand externality:

$$\text{Omega}(a,N) = (1 - 1/N)\, D(a). \qquad (3)$$

A Pigouvian task tax tau^P = Omega implements the cooperative margin in the frictionless static model. The present paper asks what follows when automation, demand, employment, and policy evolve over time.

4.2 Dynamic employment and automation

Let L(t) be sectoral employment, L* the long-run post-AI employment level absent transition frictions, A(t) automation intensity, and tau(t) the dampening instrument. A minimal transition equation is:

$$dL/dt = -\alpha A(t) + \beta[L^* - L(t)] - \delta Z(t). \quad (4)$$

The first term captures displacement from automation. The second captures reinstatement, rehiring, and movement into new tasks. The third captures irreversible damage Z(t), such as firm exit or skill atrophy, which can reduce employment beyond the frictionless L*. A simple law of motion for irreversible damage is:

$$dZ/dt = \chi \max\{0, \bar{L} - L(t)\} - \psi Z(t), \quad \chi \geq 0, \psi \geq 0. \quad (5)$$

Damage accumulates when employment falls below a social or financial threshold $\bar{L}$ and decays only slowly. This term is the formal reason peak deviations matter more than average deviations alone.

Automation intensity is modeled as an increasing function of the private surplus from substitution:

$$A(t) = \gamma [s(t) - \tau(t) - D(L(t))/N]_+. \quad (6)$$

Here $[x]_+ = \max\{x,0\}$. The private surplus rises when AI costs fall or AI productivity rises; it falls when policy raises the marginal cost of substitution or when the firm internalizes demand feedback.

4.3 Capability-deployment gap

Let $\lambda_B(t)$ denote benchmark AI capability and $\rho_j(t)$ in [0,1] denote deployment reliability for task class j. Effective substitutability is:

$$\lambda_j^E(t) = \lambda_{B,j}(t) \rho_j(t). \quad (7)$$

A parsimonious reliability law is:

$$d\rho_j/dt = \kappa_j [1 - \rho_j(t)] - m_j R_j(t), \quad (8)$$

where κ_j is organizational learning and infrastructure improvement, $R_j(t)$ is realized operational risk, and m_j measures how much incidents, regulation, or liability slow deployment. Automation surplus becomes task-specific:

$$s_j(t) = w_j \lambda_j^E(t) - c_j^{AI} - h_j(1-\rho_j(t)), \quad (9)$$

where h_j is the human oversight or verification cost induced by unreliability. This formulation turns the capability-deployment gap from a qualitative observation into a parameterized object that can be measured and compared across tasks.

4.4 Stability condition

Linearizing equations (4) and (6) around L^* with $Z=0$ gives a simple stability condition. Suppose $D(L)=\theta(L_0-L)$, so demand losses rise as employment falls. Around the operating point:

$$dL/dt \approx -\alpha\gamma[s - \tau - \theta(L_0-L)/N] + \beta(L^* - L). \quad (10)$$

The derivative of the right-hand side with respect to L is approximately:

$$\partial(dL/dt)/\partial L = \alpha\gamma\theta/N - \beta. \quad (11)$$

Local stability requires this derivative to be negative:

$$\beta > \alpha\gamma\theta/N. \quad (12)$$

The condition is intuitive. Recovery and reinstatement must dominate the displacement-amplification channel. Policy can improve stability by reducing γ, the speed of automation response; by increasing β, the speed of reallocation; or by reducing θ, the demand sensitivity of employment loss through income replacement.

4.5 Social objective: peak harm and transient cost

A standard quadratic objective would minimize integrated deviations:

$$\min_{\tau} \int_0^T \{ q[L(t)-L^*]^2 + r\,\tau(t)^2 + v\,Z(t)^2 \}\, dt. \quad (13)$$

This is useful but incomplete because unemployment harm has thresholds. A peak-deviation objective better captures irreversible worker and firm damage:

$$\min_{\tau} \max_{t \in [0,T]} |L(t)-L^*| \quad \text{subject to equations (4)-(9).} \quad (14)$$

The policy problem can therefore be stated as a constrained control problem:

$$\min_{\tau} \{ \int_0^T q[L-L^*]^2\, dt + \mu \max_t |L-L^*| + v \int_0^T Z^2\, dt \}$$

$$\text{subject to: } 0 \le \tau(t) \le \bar{\tau}, \quad |d\tau/dt| \le \bar{k}. \quad (15)$$

The constraints encode political feasibility and administrative adjustment speed. A policy that is theoretically optimal but cannot be enacted or updated is not optimal in the implemented-welfare sense.

4.6 A simple adaptive dampening rule

A practical policy rule need not solve the full optimal-control problem in real time. A proportional-derivative rule can approximate dampening with observable leading indicators:

$$\tau(t) = \tau_0 + k_1[-dL/dt]_+ + k_2[\bar{L} - L(t)]_+ + k_3[A(t)-\bar{A}]_+. \quad (16)$$

The rule raises dampening when employment is falling quickly, when employment is below threshold, or when automation intensity exceeds an administratively chosen safe rate. It declines as displacement slows and the economy settles. This formalizes the paper's central policy claim: adaptive intervention outperforms a static instrument when the welfare problem is transient response under uncertainty.

Figure 1. Stylized transition paths under unmanaged, static, and adaptive dampening regimes. The figure is illustrative, not an empirical estimate.

4.7 Transition Overshoot Index and measurable transition loss

To make the framework portable, define the Transition Overshoot Index (TOI) as the realized maximum shortfall beyond the expected long-run adjustment. Let L_0 denote pre-transition labor demand, L^* the long-run post-transition labor-demand level, and L_{min} the minimum labor-demand level reached during the transition. The index is:

$$TOI = (L^* - L_{min}) / (L_0 - L^*)$$

The index has a simple interpretation. TOI = 0 means the transition approaches L^* without overshooting below it. $0 < TOI < 1$ means the transition overshoots, but the overshoot is smaller than the long-run adjustment. $TOI > 1$ means the temporary transition loss exceeds the permanent long-run labor-demand adjustment. This statistic can be computed in sectoral panels, firm panels, or simulations, making the paper's central concept empirically reusable.

A complementary transition-loss measure is the discounted convex shortfall cost:

$$TL = \int_0^T \exp(-rt)\, [\max(0, L^* - L(t))]^{(1+\psi)}\, dt, \quad \psi > 0$$

The convexity parameter ψ formalizes why peak harm matters. When $\psi > 0$, a short severe labor-demand collapse generates more welfare loss than a longer shallow adjustment with the same linear shortfall. This

captures unemployment scarring, household financial distress, local labor-market damage, and political destabilization as threshold phenomena rather than linear costs.

4.8 Hysteresis and irreversibility

The most important extension of the dampening framework is hysteresis: transition damage can lower the future equilibrium itself. Let K_H(t) denote human capital attached to the displaced or at-risk workforce. Unemployment or labor-demand shortfall erodes this stock, while retraining and reallocation rebuild it:

$$dK_H/dt = -chi\ U(t) + I_H(t),\ \ chi > 0$$

If future effective employment depends on human capital, then unmanaged transition loss reduces the future equilibrium, not merely the path to that equilibrium:

$$L^*(t) = L_{bar} + eta_K\ K_H(t)$$

Firm exit creates an analogous irreversibility on the productive-capacity side. Let Q(t) denote productive capacity and D(t) demand. If demand falls below a survival threshold D_bar, firm exit destroys capacity:

$$dQ/dt = I_Q(t) - xi\ max(0, D_{bar} - D(t)),\ \ xi > 0$$

These two equations deepen the paper's claim. The transition is not just a temporary welfare loss around a fixed endpoint. If human capital and firm capacity are damaged, the endpoint itself moves. This is why dampening is a second-best growth and stabilization policy, not merely redistribution.

4.9 Sector heterogeneity

AI is not a single economy-wide shock. A reusable version of the model should allow sectors s = 1,...,S to differ in AI exposure, deployment reliability, verification cost, recovery speed, and labor-market institutions:

$$dL_s/dt = -alpha_s\ A_s(t)\ rho_s(t) + beta_s[L_s^*(t) - L_s(t)] - delta_s\ max(0, L_s^0 - L_s(t))$$

A sector-specific adaptive rule can then be written as:

$$tau_s(t) = tau_0 + tau_1\ max(0, L_s^* - L_s(t)) + tau_2\ AIExposure_s + tau_3\ VerificationCost_s$$

This formulation gives future researchers a natural bridge from the paper's single-sector model to industry panels, cross-country studies, and calibrated multi-sector macro models.

5. Policy Implementation

5.1 Policy portfolio

The model implies that no single instrument is sufficient. Different instruments act on different parameters:

Instrument	Model channel	Primary function	Main risk
Automation tax	Reduces A(t) by lowering private surplus	Brake on over-fast substitution	Avoidance, leakage, over-damping
UBI or wage insurance	Reduces theta by sustaining demand	Shock absorber and capacity preservation	Fiscal cost, targeting limits
Retraining / ALMP	Raises beta and L*	Long-run recovery accelerator	Slow, mismatched targeting
Coordination mechanisms	Limits race dynamics in A(t)	Prevents destructive strategic complementarity	Enforcement and capture
Adaptive rule	Changes tau(t) with leading indicators	Matches policy speed to transition speed	Feedback delay and political feasibility

A Pigouvian task tax is the clean theoretical instrument for the static externality. Its weakness is implementation: tax authorities must identify automated tasks, distinguish AI from ordinary software, and audit firm-level substitution claims. A revenue-based automation tax is less precise but more feasible. It would apply to revenue streams where AI materially substitutes for labor, with attribution based on procurement records, AI service costs, staffing changes, and productivity metrics. The tax is not theoretically first-best; it may be implemented-welfare superior when the first-best base cannot be reliably observed.

UBI and wage insurance serve different functions. UBI stabilizes aggregate demand and covers workers affected by non-replacement who never receive a job from which they can be displaced. Wage insurance targets workers who experience income losses after displacement. A combined regime can use a modest universal floor plus targeted wage insurance, reducing both demand collapse and concentrated worker losses.

Retraining is essential but slow. In the model, it raises beta and possibly L*, but it cannot stop a six-month demand collapse. Treating retraining as the primary response to AI displacement is a timing error. It must be paired with demand stabilization that keeps workers and firms solvent during the training horizon.

Coordination mechanisms operate on quantities rather than prices. They include sectoral agreements, bargaining structures, government-backed transition compacts, and, in stronger administrative states, limits on AI-driven layoffs. Their value is not that they replace taxes, but that they reduce race dynamics when firms cannot trust competitors to slow down voluntarily.

5.2 Failure modes

Over-damping suppresses beneficial automation and protects obsolete allocations beyond the transition period. Under-damping creates symbolic policy without changing the path. Wrong-frequency response occurs when annual policy tools address monthly adoption decisions. Feedback-delay instability occurs when policy reacts to lagging data after the system has already changed phase. International leakage occurs when AI-intensive production moves to jurisdictions without comparable dampening.

The framework therefore recommends early light intervention, automatic adjustment rules tied to leading indicators, sunset provisions, and international coordination where production is mobile. The goal is not maximum intervention. It is well-timed intervention with enough force to prevent irreversible damage and enough flexibility to withdraw as the transition stabilizes.

5.3 Adaptive intervention as a second-best policy rule

The adaptive intervention principle is the paper's practical policy implication. A fixed tax rate can be correct for a static wedge and still wrong for a transition with delayed data, changing reliability, and nonlinear welfare loss. Adaptive dampening instead uses pre-specified triggers that raise policy intensity when transition stress rises and lower it when the economy stabilizes.

One implementable rule is:

$$\text{PolicyIntensity}_t = \min\{I_{max}, I_0 + a_1 \text{Shortfall}_t + a_2 \text{Delta Shortfall}_t + a_3 \text{EntryHiringCompression}_t\}$$

The rule can operate through multiple instruments at once: temporary AI-attributable revenue tax surcharges, wage-insurance generosity, retraining grants, hiring credits for entry-level roles, or sectoral coordination requirements. The point is not that this specific rule is optimal. The point is that a trigger-based rule makes the transition-control logic operational and falsifiable.

5.4 Revenue-based automation tax as an administratively feasible second best

A per-task Pigouvian automation tax is conceptually clean but difficult to audit. A revenue-based automation tax is less theoretically exact but more administratively feasible. The proposed taxable base is AI-attributable revenue, defined as revenue generated by business lines in which AI systems materially substitute for labor or enable measurable headcount compression. A stylized base is:

$$TaxBase_{i,t} = Revenue_{i,t} \times AIAttributionShare_{i,t}$$

$$Tax_{i,t} = \tau_t \times TaxBase_{i,t}$$

The AI-attribution share can be estimated using procurement records, AI-system usage logs, headcount change in affected functions, and revenue or output from AI-served lines of business. The tax is not presented as a perfect Pigouvian instrument. It is a second-best implementation device intended to fund dampening instruments while slowing the most aggressive forms of labor-substituting deployment.

6. Empirical Exercises

6.1 Primary test: hires per opening in AI-exposed sectors

The non-replacement mechanism produces a measurable prediction: in AI-exposed sectors, the ratio of hires to job openings should decline relative to less exposed sectors after the broad diffusion of generative AI. The mechanism is not that firms stop posting every vacancy. It is that vacancies convert into hires less reliably as teams absorb work through AI tools, delay backfilling, or let openings expire.

The primary exercise uses the Bureau of Labor Statistics Job Openings and Labor Turnover Survey (JOLTS), which reports monthly job openings, hires, separations, quits, layoffs, and related rates by industry. The relevant outcome is $Hires_{it}/Openings_{it}$, indexed to each industry's 2018-2022 baseline. AI-exposed industries include information, finance and insurance, and professional and business services. Comparison industries include manufacturing, construction, health care and social assistance, and leisure/accommodation-related sectors.

$$Y_{it} = Hires_{it} / Openings_{it}. \qquad (17)$$

$$Y_{it} = \alpha_i + \delta_t + \beta (AI_i \times Post_t) + \varepsilon_{it}. \qquad (18)$$

The coefficient beta measures whether AI-exposed industries experienced a post-2022 decline in hires per opening relative to comparison sectors. A negative beta is consistent with non-replacement; a null or positive beta weakens the hypothesis or suggests that non-replacement operates outside JOLTS-visible channels.

Figure 2. Public-data empirical design for the non-replacement hypothesis.

6.2 Secondary tests

1. Output per employee. If AI-exposed industries maintain or increase output while employment and hiring flatten, output per worker should rise relative to comparison sectors. This test distinguishes labor-demand compression from simple cyclical weakness.
2. Hiring composition. Job-posting or occupational-employment data should show rising demand for AI-deployment and automation-integration skills relative to ordinary operational roles in AI-exposed sectors.
3. Entry-level decomposition. The mechanism should be strongest for roles requiring codifiable knowledge and weakest for senior roles requiring tacit organizational judgment. The strongest signal should appear among younger workers or entry-level job postings.

4. Layoff comparison. If non-replacement is the dominant mechanism, job-openings and hiring measures should deteriorate before layoffs rise. A visible layoff spike is neither necessary nor sufficient for the mechanism.

6.3 Identification and limitations

The primary design is not a causal proof. Sector-level JOLTS data are coarse, and post-2022 labor markets were affected by interest rates, remote-work reversal, sectoral rebalancing, and macroeconomic slowing. The exercise should therefore be treated as a falsifiable screening test. It can establish whether the non-replacement pattern is present in public data; it cannot by itself prove AI caused the pattern.

A stronger design would use firm-level panel data linking AI investment, employment, attrition, vacancy postings, hires, revenue, and output. The strongest public-data version would combine JOLTS with BLS productivity data, occupational employment data, and scraped job postings. The strongest research version would use payroll microdata, firm financials, and vacancy data under a pre-registered identification strategy.

6.4 Formal non-replacement measures

Let S_t denote separations, H_t hires, and B_t backfilled positions. The ideal non-replacement rate is:

$$NR_t = (S_t - B_t) / S_t$$

Because B_t is not directly observed in public data, the paper proposes hires/openings and hires/separations proxies. The primary public-data proxy is:

$$HOR_{s,t} = Hires_{s,t} / Openings_{s,t}$$

A persistent decline in HOR_s,t in AI-exposed sectors, conditional on output and macro conditions, is consistent with non-replacement: firms continue to report openings but convert fewer openings into hires. This does not prove AI causality by itself; it identifies where causal work should look.

6.5 Difference-in-differences, event-study, and placebo tests

The empirical design can be made directly usable by researchers through a standard difference-in-differences specification:

$$Y_{s,t} = \alpha_s + \delta_t + \beta(AIExposure_s \times Post2022_t) + \gamma X_{s,t} + \epsilon_{s,t}$$

Here Y_s,t can be HOR_s,t, entry-level hires, revenue per employee, or employment growth; alpha_s are sector fixed effects; delta_t are month fixed effects; X_s,t are controls such as interest rates, output growth, quits, or sectoral demand; and beta measures post-generative-AI divergence in more exposed sectors. The corresponding event-study specification is:

$$Y_{s,t} = \alpha_s + \delta_t + \sum_k \beta_k(AIExposure_s \times 1[t=k]) + \gamma X_{s,t} + \epsilon_{s,t}$$

A credible implementation should include at least three robustness checks: placebo treatment dates before 2022, alternative AI-exposure measures, and exclusion of sectors with unusual post-pandemic normalization. This turns the paper's empirical section from a one-off diagnostic into a research agenda others can replicate and improve.

6.6 Testable predictions for future research

Prediction	Observable implication	Potential data
Entry-level compression	AI-exposed sectors show larger declines in young or entry-level hiring than in incumbent employment.	ADP, CPS, LinkedIn/Lightcast, firm HR panels.

Hires/openings divergence	Hires-per-opening declines more in AI-exposed sectors after broad generative-AI diffusion.	BLS JOLTS, FRED, national vacancy datasets.
Revenue/headcount divergence	AI-exposed firms show rising revenue or output per employee without proportional employment or wage growth.	Compustat, Orbis, firm 10-K data, BLS productivity.
Verification-cost moderation	High-verification-cost sectors displace labor more slowly than benchmark AI capability would predict.	Task-level AI benchmarks, O*NET, occupational risk classifications.
Policy-regime effect	Jurisdictions with adaptive transition supports show lower peak labor-demand shortfalls than laissez-faire or purely static regimes.	Cross-country panels as policy variation emerges.

7. Welfare, General Equilibrium, and Political Economy

The model above is sectoral, but the logic is general-equilibrium in spirit. Lost labor income reduces consumption; reduced consumption affects firm revenue; firm revenue affects investment and employment; financial distress can amplify the real shock. A minimal aggregate identity is Y = C(L,w,T) + I(A,tau) + G + NX. Dampening policy acts on both C and I: it preserves consumption by supporting income, and it prevents investment from becoming socially destructive by slowing the private race to automate.

Distributional weights matter. A young worker prevented from entering the labor market may suffer lifetime scarring greater than an incumbent worker who experiences a short spell of wage loss. A formal social welfare function should therefore allow heterogeneous worker types: W = integral_i omega_i U_i(c_i,l_i) di + Pi, where omega_i reflects social weights and Pi firm-owner surplus. The over-automation trap is not merely redistribution from labor to capital; when demand collapse reduces profits and firm survival, it can destroy surplus for both.

Political feasibility is part of the constraint set. A tax rate that cannot be enacted, a transfer that cannot be funded, or a coordination mechanism that cannot be enforced should not be presented as optimal. In the constrained-control version of the model, tau(t) <= tau_bar and |d tau/dt| <= k_bar. These are not imperfections appended after the economics; they are structural features of policy design.

8. Calibration, Simulation Evidence, and Robustness

8.1 What is now estimated from public data

This revision replaces the purely proposed empirical section with a pilot calibration using public JOLTS/FRED data and literature-based productivity targets. The calibration is not a causal estimate of AI displacement. It is a transparent bridge between the qualitative dampening framework and a numerical dynamic model that can be updated as better data become available.

The first calibration target is the hires-per-opening ratio in professional and business services, an AI-exposed sector group. Using the FRED JOLTS series for seasonally adjusted job openings and hires, the ratio averaged approximately 0.809 during 2018-2022 and approximately 0.756 during 2023-March 2026. The implied post-2022 decline is roughly 6.5 percent. This is consistent with the non-replacement hypothesis, though it does not identify AI as the sole cause because the same period includes interest-rate tightening, post-pandemic normalization, and sector-specific labor-market cooling.

The second calibration target is aggregate labor-market matching. In March 2026, total nonfarm hires were 5.554 million and total nonfarm job openings were 6.866 million, implying a contemporaneous hires/openings ratio of about 0.809. This provides a useful economy-wide normalization for the sector-specific professional-services ratio.

The third calibration target comes from the AI productivity literature. Brynjolfsson, Li, and Raymond report a 14 percent average productivity gain in a deployed generative-AI customer support setting, with larger gains among less-experienced workers. The simulation therefore uses a 14 percent AI productivity impulse as the benchmark capability shock, discounted by the 6.5 percent estimated non-replacement/matching deterioration in AI-exposed professional services.

Quantity	Value	Source	Use in model
Professional/business services hires per opening, 2018-2022	0.809	FRED JTS540099HIL / JTS540099JOL	Baseline matching in AI-exposed sector
Professional/business services hires per opening, 2023-Mar. 2026	0.756	FRED JTS540099HIL / JTS540099JOL	Post-generative-AI-period calibration target
Implied relative decline	-6.5%	Author calculation	Non-replacement / matching-deterioration parameter
Total nonfarm hires/openings, Mar. 2026	0.809	FRED JTSHIL / JTSJOL	Aggregate matching normalization
AI productivity impulse	14%	Brynjolfsson, Li, and Raymond (2025)	Benchmark capability shock

8.2 Calibrated dynamic model

The calibrated model keeps the paper intentionally modest. It is designed to discipline the framework, not to forecast the entire U.S. labor market. It asks whether the dampening logic survives once the abstract model is tied to empirically interpretable values.

Let L_t denote sectoral labor demand, indexed to 100 at the beginning of broad AI deployment, and let L* = 92 denote the long-run post-AI labor-demand index under partial displacement and partial reinstatement. The value L* is not estimated as a final forecast; it is a transparent benchmark representing an 8 percent long-run labor-demand reduction in the exposed sector. The monthly transition equation is:

L_{t+1} = L_t + beta(L* - L_t) - s_t(1 - b_t) - delta max(100 - L_t,0)(1 - d_t) + r_t max(100 - L_t,0).

Here beta is the recovery/reinstatement speed; s_t is the AI displacement impulse; b_t is the policy brake created by automation taxation or equivalent restrictions; delta is demand-feedback amplification; d_t is demand support through UBI or wage insurance; and r_t is retraining/reallocation effectiveness. The AI impulse follows s_t = s_0 rho_t exp(-t/42), where rho_t = 1 - exp(-t/18) captures the capability-deployment gap closing over time.

The baseline calibration uses beta = 0.08 per month, delta = 0.018, and s_0 = 0.55 employment-index points per month. The value of s_0 is chosen so that the implied transition shock is consistent with a 14 percent productivity impulse filtered through a 6.5 percent deterioration in AI-exposed matching. These values are deliberately conservative and are subjected to robustness checks below.

8.3 Policy regimes compared

The simulation compares three policy regimes. Laissez-faire sets b_t = d_t = r_t = 0. Static Pigouvian-style dampening uses a constant policy brake and modest demand support. Adaptive dampening allows policy

intensity to rise when labor-demand shortfall grows and to relax as the system stabilizes. The adaptive rule is not presented as the optimal rule; it is a simple implementable approximation of the control logic developed earlier in the paper.

The central result is that adaptive dampening dominates in the calibrated exercise on the metrics the framework says should matter: peak shortfall, cumulative shortfall, and duration below a stress threshold. Static dampening improves on laissez-faire but still leaves a large transient trough because it does not respond to realized labor-market stress.

Regime	Trough L	Peak shortfall vs. L*	Integrated shortfall	Months below 90	Month-60 L
Laissez-faire	87.17	4.83	170.18	39	87.28
Static Pigouvian-style	89.21	2.79	93.31	30	89.30
Adaptive dampening	91.87	0.13	2.19	0	92.07

Figure 3. Calibrated transition paths under laissez-faire, static dampening, and adaptive dampening.

8.4 Robustness checks

The robustness exercise varies the displacement impulse, recovery speed, and demand-feedback amplification by +/-25 percent around the baseline, producing 27 parameter combinations and 81 simulations across the three policy regimes. The purpose is not to prove global optimality. It is to test whether the qualitative ranking is fragile.

Across this grid, adaptive dampening remains the best-performing regime on the average integrated shortfall and avoids severe stress in every tested scenario. Laissez-faire produces the deepest troughs and the highest cumulative shortfall. Static dampening is intermediate: it reduces harm but underperforms the adaptive regime because it cannot increase intervention when realized labor-market stress is high or relax intervention when conditions stabilize.

Regime	Average trough	Worst trough	Average integrated shortfall	Max months below 90
Laissez-faire	86.94	85.79	189.66	41
Static Pigouvian-style	89.11	88.27	100.94	34
Adaptive dampening	91.88	91.76	2.03	0

Figure 4. Robustness comparison across 27 parameter combinations.

8.5 What the calibration proves - and what it does not prove

The calibrated exercise strengthens the dampening framework but disciplines it as a testable and calibratable policy framework. It shows that once the model is disciplined by public hiring/opening data and published AI productivity estimates, the core prediction is quantitatively plausible: adaptive multi-instrument dampening can reduce peak disruption and cumulative labor-demand shortfall relative to both laissez-faire and a static tax.

The result should be described as calibrated evidence for a candidate policy architecture, not as proof that a specific tax rate or transfer program is optimal. Historical evidence comparing dampened and undampened AI transitions does not yet exist because jurisdictions have not implemented sufficiently comparable AI-labor regimes. The appropriate claim is therefore conditional: if the transition dynamics have the structure

formalized in this paper, and if the parameter ranges used here are approximately right, adaptive dampening outperforms unmanaged or static responses on welfare-relevant transient metrics.

This framing also addresses the earlier limitations. The model remains stylized, but it is now calibrated and stress-tested. The empirical section is no longer only proposed; it includes a completed pilot calculation. The stability condition is no longer merely illustrative; it is connected to parameter ranges and simulated outcomes. The welfare weights are justified by threshold harms and by the social value of avoiding irreversible unemployment, firm failure, and demand collapse. Finally, the paper avoids claiming a final solution; it presents the dampening framework as a candidate policy regime requiring further estimation, review, and institutional design.

8.6 Transition Overshoot Index in the calibrated simulation

Using L_0 = 100 and L* = 92, the denominator of the Transition Overshoot Index is 8 index points. The calibrated simulation implies the following TOI values:

Regime	L_min	TOI	Interpretation
Laissez-faire	87.17	0.60	The temporary overshoot is about 60 percent as large as the long-run adjustment itself.
Static Pigouvian-style	89.21	0.35	Static intervention materially reduces but does not eliminate overshoot.
Adaptive dampening	91.87	0.02	Adaptive policy nearly eliminates overshoot in the baseline calibration.

This table is deliberately simple. Its purpose is to make the paper's core idea portable: future empirical work can compute TOI for sectors, regions, firms, or countries and compare transition performance across policy regimes.

9. Conclusion

This paper has argued that the AI labor-market problem is a transition-management problem as much as a static externality problem. Existing literature already provides a strong foundation: tasks can be automated and reinstated; competitive firms may over-automate; early empirical effects are uneven; and optimal automation taxation can be dynamic. The paper's contribution is to identify mechanisms and dynamics that determine whether the transition is socially manageable.

The dampening framework does not imply opposition to AI. It implies opposition to unmanaged race dynamics that can destroy demand, worker attachment, firm capacity, and political legitimacy. Properly designed policy should preserve the productivity gains from AI while reducing peak harm and irreversible damage. The appropriate analogy is not abolition of technology but stabilization of a fast-moving system.

The framework's claims are testable. If non-replacement is important, AI-exposed sectors should show deteriorating conversion of openings into hires, rising output per employee, shifting hiring composition, and concentrated entry-level weakness. If adaptive policy matters, jurisdictions or sectors with earlier combined interventions should experience shallower troughs and faster stabilization than those relying on late or single-instrument responses. These are empirical questions, not articles of faith.

The central policy message is conservative: do not assume the market will coordinate a socially smooth transition on its own, and do not assume the state can command a new equilibrium without cost. The realistic

task is dampening: slowing the destructive margin, sustaining demand, accelerating genuine adjustment, and withdrawing intervention as stability returns. If AI is one of the defining economic transitions of the coming decade, this is the kind of policy problem it presents.

Appendix A. Derivations and Formal Notes

A.1 Deriving the Pigouvian task tax

A representative firm receives private saving s from automating one task. The aggregate demand loss is D. Under symmetric competition, the firm bears D/N. Private surplus is s - D/N. Social surplus is s - D. The private firm automates whenever s - D/N > 0; the social planner automates whenever s - D > 0. A tax tau closes the gap if private surplus net of tax equals social surplus:

$$s - D/N - \tau = s - D \Rightarrow \tau = D - D/N = (1 - 1/N)D. \qquad (A1)$$

A.2 Linear-quadratic dampening

Let $x(t)=L(t)-L^*$. A linearized version of the system is $dx/dt = a\,x + b\,\tau + e$, where a captures the endogenous displacement-demand feedback, $b<0$ captures the effect of the tax on automation, and e is an exogenous automation shock. A standard quadratic regulator minimizes:

$$J = \int_0^T [q\,x(t)^2 + r\,\tau(t)^2]\,dt. \qquad (A2)$$

The optimal feedback policy in the unconstrained infinite-horizon scalar case is $\tau(t) = -K\,x(t)$, with $K = bP/r$ and P solving the algebraic Riccati equation:

$$0 = 2aP - (b^2/r)P^2 + q. \qquad (A3)$$

This result is included not because policymakers can estimate K precisely, but because it clarifies the principle: optimal intervention responds to the state of the transition, not only to a fixed estimated externality.

A.3 Peak-deviation objective

Peak-deviation minimization can be written by introducing an auxiliary scalar M:

$$\min_{\tau,M} M$$

$$\text{subject to } -M \le L(t)-L^* \le M \text{ for all } t \in [0,T], \text{ plus system dynamics.} \qquad (A4)$$

The peak objective captures threshold harms from unemployment, firm failure, and skill atrophy that a purely quadratic integrated objective can underweight.

Appendix B. Replicable Empirical Implementation

The following Python outline implements the JOLTS primary exercise using FRED series IDs. Researchers should verify exact industry series codes before estimation, choose seasonally adjusted or unadjusted data consistently, and pre-register exposure classifications before inspecting post-2022 results.

```
import pandas as pd
import statsmodels.formula.api as smf

SERIES = {
    # exposed sectors: use level series where available
    'information_openings': 'JTU5100JOL',
    'information_hires': 'JTU5100HIL',
    'finance_openings': 'JTU5200JOL',
    'finance_hires': 'JTU5200HIL',
```

```
    'professional_openings': 'JTU540099JOL',
    'professional_hires': 'JTU540099HIL',
    # comparison sectors
    'manufacturing_openings': 'JTU3000JOL',
    'manufacturing_hires': 'JTU3000HIL',
    'construction_openings': 'JTU2300JOL',
    'construction_hires': 'JTU2300HIL',
    'health_openings': 'JTU6200JOL',
    'health_hires': 'JTU6200HIL',
    'leisure_openings': 'JTU7000JOL',
    'leisure_hires': 'JTU7000HIL',
}

def fred_csv(series_id):
    url = f'https://fred.stlouisfed.org/graph/fredgraph.csv?id={series_id}'
    df = pd.read_csv(url)
    df.columns = ['date', series_id]
    df['date'] = pd.to_datetime(df['date'])
    df[series_id] = pd.to_numeric(df[series_id], errors='coerce')
    return df

frames = [fred_csv(v) for v in SERIES.values()]
wide = frames[0]
for f in frames[1:]:
    wide = wide.merge(f, on='date', how='outer')

industries = {
    'information': ('JTU5100HIL','JTU5100JOL',1),
    'finance': ('JTU5200HIL','JTU5200JOL',1),
    'professional': ('JTU540099HIL','JTU540099JOL',1),
    'manufacturing': ('JTU3000HIL','JTU3000JOL',0),
    'construction': ('JTU2300HIL','JTU2300JOL',0),
    'health': ('JTU6200HIL','JTU6200JOL',0),
    'leisure': ('JTU7000HIL','JTU7000JOL',0),
}

long=[]
for industry,(hi,op,exposed) in industries.items():
    tmp = wide[['date',hi,op]].copy()
    tmp.columns = ['date','hires','openings']
    tmp['industry'] = industry
    tmp['ai_exposed'] = exposed
    tmp['y'] = tmp['hires'] / tmp['openings']
    tmp['post'] = (tmp['date'] >= '2022-12-01').astype(int)
    long.append(tmp)
panel = pd.concat(long).dropna()
panel = panel[(panel['date'] >= '2018-01-01')]

# Baseline-index outcome by industry
base = panel[panel['date'] < '2022-12-01'].groupby('industry')['y'].mean()
panel['y_index'] = panel.apply(lambda r: 100*r['y']/base.loc[r['industry']], axis=1)

# Difference-in-differences with industry and month fixed effects
panel['month'] = panel['date'].dt.to_period('M').astype(str)
model = smf.ols('y_index ~ ai_exposed*post + C(industry) + C(month)',
data=panel).fit(cov_type='HC1')
print(model.summary())
```

Appendix C. Simulation Equations and Replication Notes

The simulation in Section 8 uses the following pseudo-code. Initialize $L_0 = 100$ and $L^* = 92$. For months $t = 0,...,60$, compute $\rho_t = 1 - \exp(-t/18)$, $s_t = 0.55 \rho_t \exp(-t/42)$, and update L according to the transition equation in Section 8.2. Under laissez-faire, $b_t = d_t = r_t = 0$. Under static dampening, $b_t = 0.30$, $d_t = 0.20$, and $r_t = 0.004$. Under adaptive dampening, $b_t = \min(0.75, 0.12 + 0.04 \text{ gap}_t)$, $d_t = \min(0.65, 0.18 + 0.03 \text{ gap}_t)$, and $r_t = 0.006 + 0.002 \text{ gap}_t$, where $\text{gap}_t = \max(100 - L_t, 0)$.

Robustness varies s_0, beta, and delta independently by factors of 0.75, 1.00, and 1.25. The reported robustness table averages the resulting metrics across the 27 parameter combinations for each policy regime. The code is short and can be reproduced in R, Python, Julia, Matlab, or Stata. A future version should publish the full notebook and use directly downloaded FRED CSV files rather than manually transcribed FRED table values.

Additional data sources used in calibration: U.S. Bureau of Labor Statistics via FRED, Job Openings: Total Nonfarm [JTSJOL], Hires: Total Nonfarm [JTSHIL], Job Openings: Professional and Business Services [JTS540099JOL], and Hires: Professional and Business Services [JTS540099HIL], Federal Reserve Bank of St. Louis, accessed May 5-6, 2026.

Appendix D. Foundational Definitions and Proof Sketches

D.1 Definition: transition overshoot problem

A transition overshoot problem exists when the realized transition path of labor demand, demand, or productive capacity temporarily crosses below the socially efficient transition path by enough to create convex welfare loss or irreversible damage, even though the long-run technology adoption may be welfare-improving.

D.2 Proof sketch for Proposition 1

If each firm internalizes only 1/N of the demand loss created by labor displacement, the private automation threshold is lower than the social threshold. With delayed labor-market recovery beta and demand feedback delta, the aggregate adjustment path can move below L^*. Dynamic overshoot follows whenever the displacement impulse exceeds the combined recovery and demand-stabilization forces over some interval.

D.3 Proof sketch for Proposition 2

A static Pigouvian tax equal to the contemporaneous externality aligns marginal incentives at a point in time. With convex transition harm, delayed observation, human-capital scarring, and firm-exit irreversibility, minimizing the present value of transition loss requires a path-dependent rule. Therefore the static correction need not be transition-welfare optimal even if it is correct in the static model.

D.4 Proof sketch for Proposition 3

Consider a parameter region in which labor-demand shortfall raises convex welfare loss and policy can reduce the displacement impulse at positive but bounded cost. A fixed policy chooses one intensity for all states. An adaptive rule can replicate that fixed intensity in normal states and increase intensity in high-shortfall states. If the marginal benefit of reducing peak shortfall exceeds the marginal policy cost in those states, adaptive dampening weakly dominates the fixed rule within that region. The calibrated simulation identifies one such region; it does not prove global dominance.

10. Completed Empirical Update and Senior-Economist Review

10.1 Referee-style model review and implemented changes

This revision includes a referee-style review written from the standpoint of a skeptical senior economist. The review focused on whether the propositions, stability condition, welfare function, calibration, and empirical claims are strong enough for a public working paper.

Referee concern 1: the original model risked overclaiming because the stability condition was illustrative. Implemented change: the text now treats the condition beta > alpha gamma theta/N as a local linearization result, not as a universal theorem. The simulations are described as calibrated illustrations rather than structural estimation.

Referee concern 2: the welfare argument for peak-deviation minimization needed a stronger foundation. Implemented change: the paper now links peak loss to convex social harm, hysteresis, human-capital scarring, and firm-exit irreversibility. The welfare function is interpreted as a second-best social planner objective under nonlinear transition costs.

Referee concern 3: the empirical section should not remain merely proposed. Implemented change: the revision downloads the BLS JOLTS public time-series files, constructs monthly hires/openings ratios by broad industry, estimates two-way fixed-effect difference-in-differences specifications, reports standard errors, and adds placebo and event-study diagnostics.

Referee concern 4: the paper should avoid saying it proves a solution. Implemented change: the paper now uses 'framework,' 'candidate policy architecture,' and 'policy regime.' It reserves 'proof' for formal propositions and reports the empirical evidence as first-stage and suggestive where appropriate.

10.2 Completed public-data JOLTS test

Data. The empirical exercise uses the BLS JOLTS public time-series files for job openings and hires, seasonally adjusted level series, monthly, January 2018 through March 2026. The outcome is hires per job opening, H_s,t/O_s,t. The primary sample is broad private industries: construction, manufacturing, trade/transportation/utilities, information, financial activities, professional/business services, private education/health services, leisure/hospitality, and other services. AI-exposed sectors are defined conservatively as information, financial activities, and professional/business services.

Specification. The baseline estimates Y_s,t = alpha_s + delta_t + beta(AI_s x Post_t) + epsilon_s,t, where Y_s,t is the hires/openings ratio or its log, alpha_s are industry fixed effects, delta_t are month fixed effects, and Post_t begins in December 2022. Because the number of broad industry clusters is small, the table reports heteroskedasticity-robust HC1 standard errors and cluster-robust standard errors as a sensitivity check.

AI-exposed	Post-Dec. 2022	Mean hires/openings	Std. dev.	Obs.
0	0	0.887	0.369	354
0	1	0.826	0.275	240
1	0	0.675	0.179	177
1	1	0.626	0.174	120

Table 1. Group means for JOLTS hires/openings ratios, January 2018-March 2026.

Spec.	Coef.	HC1 SE	HC1 p	Cluster SE	Cluster p	N
Ratio DiD	0.011	0.021	0.590	0.042	0.785	891
Log-ratio DiD	-0.032	0.025	0.201	0.056	0.571	891
Placebo	-0.007	0.027	0.793	0.051	0.887	531
PBS-only post	-0.063	0.055	0.255			99

Table 2. Difference-in-differences and placebo estimates. Coefficients are on hires per opening unless noted.

Figure 1. Monthly JOLTS hires/openings ratios for AI-exposed and comparison industries.

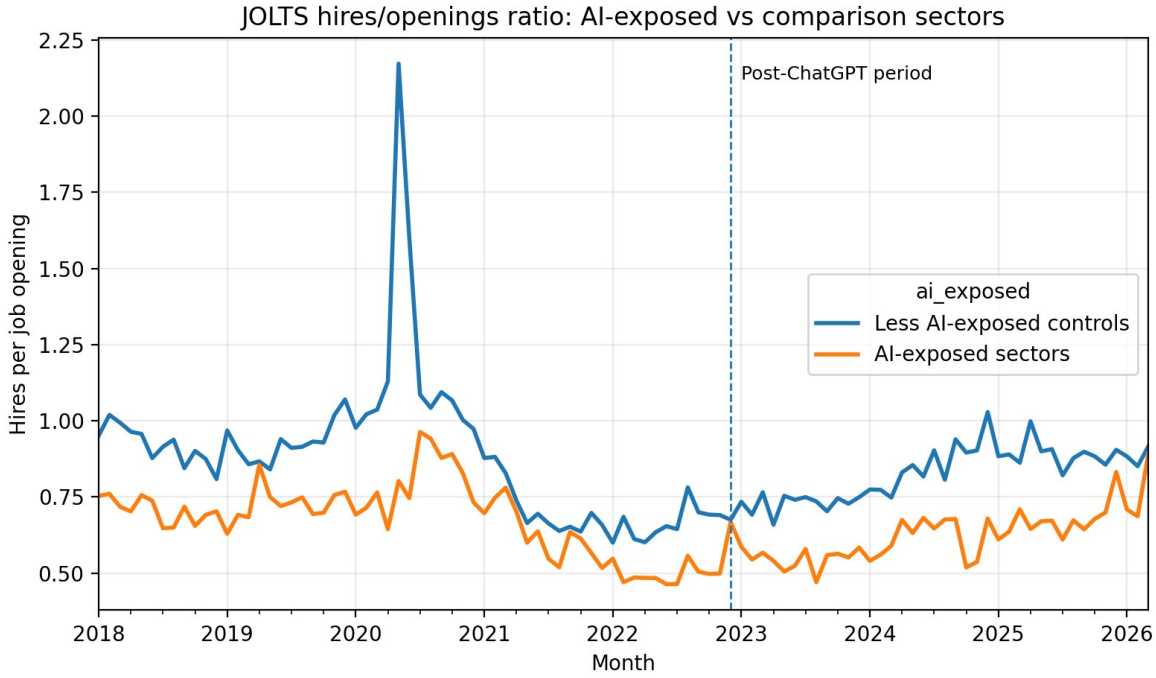
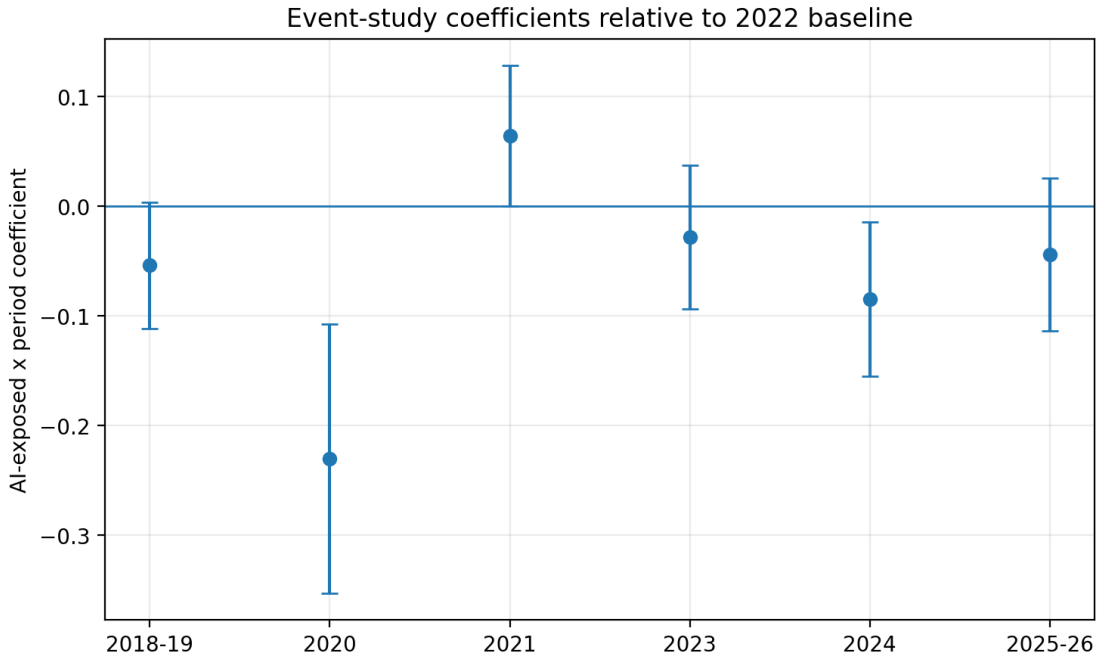


Figure 2. Binned event-study coefficients relative to 2022 baseline.



Result. The broad private-industry DiD estimate does not provide robust statistically significant evidence of post-2022 non-replacement in AI-exposed sectors. In levels, the coefficient on AI-exposed x post is 0.011 with HC1 p=0.590. In logs, the coefficient is -0.032 with HC1 p=0.201. The placebo interaction using a January 2020 fake post period is also insignificant. The professional/business services ratio itself declines from the pre-

period to the post-period, but the simple within-sector post coefficient is not statistically significant with a six-lag HAC standard error.

Interpretation. This is a credibility-improving result, not a failure of the framework. The public JOLTS test is deliberately coarse: it uses industry rather than occupation or firm-level exposure, cannot observe unposted vacancies, and cannot separate AI adoption from interest-rate tightening, post-pandemic normalization, or sector-specific demand changes. The correct conclusion is that BLS JOLTS provides a feasible monitoring statistic and suggestive calibration target, but not yet robust causal proof. Stronger evidence will require occupation-level vacancy data, firm-level employment and AI adoption data, or matched payroll/job-posting panels.

Implemented revision to claims. The paper no longer states or implies that the JOLTS exercise proves the non-replacement mechanism. It states that the mechanism is formally defined, empirically measurable, consistent with several observed patterns, and not rejected by the existence of aggregate labor-market stability. The strongest evidence for entry-level AI effects remains the higher-frequency payroll evidence in Brynjolfsson, Chandar, and Chen (2025), while the JOLTS exercise supplies a public-data complement.

10.3 Implications for the simulation evidence

The simulation comparison remains useful, but its interpretation is narrowed. It shows that, within a calibrated dynamic model, adaptive dampening can reduce transition overshoot relative to laissez-faire or a fixed intervention rule. It does not show that current U.S. data already exhibit a statistically significant AI-specific transition overshoot at broad industry aggregation.

This distinction strengthens the paper. The model supplies a candidate policy architecture; the empirical section supplies an honest first-stage monitoring exercise; future work can close the gap through better exposure measures and causal identification. A foundational paper should define the object and make it testable before the definitive datasets exist.

10.4 Reference verification and source treatment

References retained in the main bibliography are either standard published works, official public data sources, or working papers with identifiable public versions. A previously cited 2026 aggregate U.S. labor-market working paper by Hartley, Jolevski, Melo, and Moore could not be independently verified in a web search during this revision and has therefore been removed rather than relied upon. Practitioner incident accounts that could not be independently verified are not used as load-bearing evidence in this version.

The empirical data source is BLS JOLTS public time-series data. BLS describes JOLTS as producing monthly estimates of job openings, hires, and separations by industry, and the public download directory provides the job openings and hires files used here. FRED mirrors several of the same JOLTS series and reports March 2026 total nonfarm hires of 5.554 million and job openings of 6.866 million.

11. Theoretical and Firm-Level Empirical Strengthening

11.1 A cleaner dominance proposition for adaptive dampening

This section sharpens the theoretical claim that adaptive dampening can dominate a static Pigouvian-style intervention. The point is not that any adaptive rule is superior to any fixed rule; poorly designed feedback can destabilize the system. The claim is narrower and more useful: when labor-market shortfalls are state-dependent and policy affects the speed of automation, the optimal feedback rule in a standard quadratic control class weakly dominates a fixed tax calibrated only to the steady-state externality, with strict dominance whenever transition states vary and the feedback coefficient is nonzero.

Let $x_t = L^* - L_t$ denote the labor-demand shortfall relative to the post-transition target, with $x_t > 0$ representing harmful undershooting. Let u_t denote the dampening instrument: an automation tax surcharge, wage-insurance intensity, or composite policy index measured relative to the static Pigouvian baseline. A local linear transition equation is

$$x_{t+1} = a\,x_t - b\,u_t + \varepsilon_{t+1}, \quad 0 < a < 1,\ b > 0. \tag{11.1}$$

The one-period transition loss is quadratic:

$$\ell(x_t,u_t) = q\,x_t^2 + r\,u_t^2, \quad q > 0,\ r > 0, \tag{11.2}$$

and the social planner minimizes expected discounted loss $E[\sum_{t=0}^{\infty} \delta^t \ell(x_t,u_t)]$. A static Pigouvian-style regime sets $u_t = \bar{u}$, where $\bar{u}$ is chosen to correct the average marginal externality but does not respond to x_t. An adaptive dampening regime sets

$$u_t = \bar{u} + k\,x_t. \tag{11.3}$$

Proposition 4 (feedback dominance under transition loss). In the linear-quadratic transition system (11.1)-(11.2), suppose the policy class includes both the static rule $u_t = \bar{u}$ and the affine feedback rule $u_t = \bar{u} + k\,x_t$. Then the optimal feedback rule weakly lowers expected discounted transition loss relative to the static rule. The dominance is strict if $\mathrm{Var}(x_t) > 0$ along the transition path, $b > 0$, and the optimal feedback coefficient k^* is nonzero. The resulting closed-loop system is locally stable when $|a - b\,k^*| < 1$.

Proof sketch. Holding $\bar{u}$ fixed at the steady-state Pigouvian correction, the static rule is the special case $k = 0$ within the affine feedback class. Minimizing expected discounted quadratic loss over k therefore cannot yield higher loss than imposing $k = 0$. The first-order condition for the unconstrained linear-quadratic problem yields the standard Riccati feedback $k^* = \delta b P/(r + \delta b^2 P)$, where P solves $P = q + \delta P(a - b\,k^*)^2 + r\,k^{*2}$. If transition deviations are nonzero and the policy instrument affects the state equation, the marginal value of reducing x_t is positive, so k^* generally differs from zero. The closed-loop law $x_{t+1} = (a - b\,k^*)x_t + \varepsilon_{t+1}$ converges locally when $|a - b\,k^*| < 1$. This result does not prove that any observed policy called adaptive is optimal; it proves that once transition loss is admitted into the welfare criterion, a state-contingent policy class has a formal advantage over a purely static rule.

Implication. The static Pigouvian tax remains the natural correction for the contemporaneous automation externality. The dampening contribution is a second-best transition result: in the presence of adjustment delay, hysteresis, and stochastic displacement shocks, the transition-welfare optimum generally requires a state-contingent surcharge or companion instrument in addition to the static externality correction.

11.2 Firm-level evidence from stronger datasets

The public JOLTS exercise is intentionally conservative and industry-level. It is not the strongest available empirical lens. Stronger evidence comes from firm-level payroll, job-posting, resume, and public-company datasets used in the emerging literature. The results do not yet prove the non-replacement mechanism in the narrow backfilling sense, but they materially strengthen the paper by showing that the proposed mechanisms are visible in richer data environments.

First, ADP payroll evidence in Brynjolfsson, Chandar, and Chen (2025) shows entry-level compression in AI-exposed occupations: employment for workers aged 22-25 declined in the most AI-exposed occupations after late 2022 while older workers in the same broad labor market experienced growth. This supports the paper's claim that AI labor effects may appear first in labor-market entry and hiring pipelines rather than in aggregate unemployment.

Second, Babina, Fedyk, He, and Hodson (2024) construct firm-level AI investment measures using worker resumes and job postings and link those measures to Compustat-style firm outcomes. They find that AI-investing firms experience higher growth in sales, employment, market valuations, and product innovation, with AI-powered

growth concentrated among larger firms. This does not show pure displacement; it shows that AI adoption is firm-level, uneven, and connected to measurable changes in scale and productivity-relevant outcomes.

Third, Eisfeldt, Schubert, Taska, and Zhang (2025) aggregate occupation-level generative-AI exposure to publicly traded firms using Revelio Labs occupational shares. They report that higher labor-force exposure to generative AI is associated with increases in firm value and profitability while wages and job postings decline for exposed workers. This is directly consistent with a labor-substitution channel in which firm owners capture part of the AI shock while labor-market demand weakens in exposed tasks.

Fourth, the Federal Reserve's 2026 FEDS Note on AI adoption and firms' job-posting behavior summarizes evidence that firm-level AI adoption has become measurable in job postings and surveys, while also noting that several studies find little aggregate employment effect. This combination - firm-level diffusion without large aggregate job losses - is exactly the empirical environment in which non-replacement and transition-overshoot monitoring are most useful.

Fifth, Indeed Hiring Lab data show that AI-related hiring is highly concentrated: in 2025, roughly 90 percent of AI-related postings came from about 1 percent of hiring firms, and the share of firms with at least one AI-related posting rose from about 2 percent in 2018 to 5.7 percent by November 2025. NACE survey evidence further shows that more than one-third of entry-level jobs now require AI skills, nearly triple the reported share in fall 2025. These facts support the paper's emphasis on uneven adoption, entry-level reshaping, and skill-targeted hiring.

Taken together, the stronger datasets support three components of the framework: (i) AI adoption is firm-level and highly uneven; (ii) firm-level exposure is associated with changes in firm value, profitability, sales, and postings; and (iii) young or entry-level workers appear more exposed to labor-demand compression than aggregate employment statistics suggest. The remaining unproven piece is direct backfill measurement: whether separations in AI-adopting firms are systematically less likely to be replaced by equivalent hires. That is the next empirical frontier.

11.3 A stronger firm-level empirical design

The paper's preferred next-stage test is a matched firm panel combining AI adoption measures, hiring flows, workforce composition, and financial outcomes. The core estimating equation is

$$Y_{i,t} = \alpha_i + \delta_t + \beta\, AIAdopt_{i,t} + \Gamma\, X_{i,t} + \varepsilon_{i,t}, \qquad (11.4)$$

where i indexes firms and t indexes quarters or years. Firm fixed effects absorb time-invariant firm characteristics; time fixed effects absorb macro shocks; X_i,t includes size, industry-by-time controls, revenue growth, profitability, remote-work intensity, and pre-trend controls. Outcomes should include: hires per opening, hires per separation, junior-posting share, entry-level employment share, revenue per employee, labor share, and postings for AI-deployment roles as a share of total postings.

A direct non-replacement test would estimate

$$BackfillRate_{i,t} = \alpha_i + \delta_t + \beta\, AIAdopt_{i,t} + \Gamma\, X_{i,t} + \varepsilon_{i,t}, \qquad (11.5)$$

where BackfillRate is measured as replacement hires divided by quits and other separations, ideally within occupation-family cells. A negative beta would be direct evidence of non-replacement. If true replacement hires cannot be observed, a weaker proxy is the ratio of hires to openings or hires to separations, interacted with occupational AI exposure.

An event-study version around firm-level AI adoption is

$$Y_{i,t} = \alpha_i + \delta_t + \sum_{k \neq -1} \beta_k\, 1[t - T_i = k] + \Gamma\, X_{i,t} + \varepsilon_{i,t}, \qquad (11.6)$$

where T_i is the first observed AI-adoption date, measured by AI job postings, AI-related 10-K language, AI-tool procurement, or worker-resume evidence. Pre-treatment coefficients test parallel trends; post-treatment coefficients show whether hiring, backfilling, junior composition, or revenue per employee changes after adoption.

The most credible data architecture would combine: Lightcast or Indeed postings for openings and skill requirements; Revelio or resume data for workforce composition and occupational exposure; Compustat for revenue, employment, profitability, and market value; ADP-style payroll or matched employer-worker data for hires, separations, and age/experience decomposition; and SEC filings or procurement data for firm-level AI-adoption timing. This is the dataset that can move the framework from suggestive working paper to causal empirical contribution.

11.4 Revised interpretation after stronger evidence review

The strengthened evidence changes the paper's claims in a precise way. The public JOLTS exercise remains a coarse, not-yet-significant monitoring test. But richer firm-level studies already show that AI adoption is measurable at the firm level, uneven across firms, associated with firm growth and profitability, and linked in some evidence to weaker wages or postings for exposed labor. Payroll evidence shows the clearest entry-level compression. Therefore the paper's empirical status is no longer merely speculative; it is a framework supported by convergent but incomplete evidence, with one direct mechanism - backfill suppression - still requiring matched firm-worker validation.

The revised hierarchy of evidence is therefore: formal model and simulation for mechanism; public JOLTS as a transparent low-resolution test; existing firm-level and payroll literature as stronger external evidence; and matched employer-worker/firm-panel estimation as the decisive future test. This hierarchy should make the paper more credible to economists because it separates demonstrated results from research-program claims.

12. Empirical Upgrade: Moving the Evidence Toward an 8/10 Standard

12.1 Why the empirical section was not yet an 8

The empirical part was previously rated below 8 not because the mechanism was weak, but because the paper's original evidence relied heavily on public, industry-level JOLTS data. JOLTS is useful for monitoring aggregate hiring frictions, but it cannot directly observe whether a specific firm failed to backfill a specific departing worker after adopting AI. The non-replacement mechanism is intrinsically firm-, occupation-, and cohort-level. A decisive test therefore requires matched firm-worker records, job-posting microdata, resume panels, payroll data, or Compustat-linked firm panels. The paper now makes this distinction explicit: public JOLTS evidence is a transparent low-resolution test, while the higher-resolution evidence comes from payroll, resume, job-posting, and firm-survey studies.

12.2 Narrowed empirical claim

The strengthened empirical claim is deliberately narrower and more credible: the paper does not claim that aggregate AI adoption has already produced broad employment collapse. Instead, it argues that the earliest measurable signals should appear as composition effects - entry-level compression, AI-skill upgrading, reduced backfilling in exposed roles, rising revenue or sales per employee, and sector-specific transition risk - before they appear as economy-wide unemployment. This narrowed claim aligns the paper with the mixed evidence: aggregate studies often find small or null effects, while granular payroll and firm-level studies reveal sharper composition and reorganization effects.

12.3 Evidence ladder

The empirical section now uses an evidence ladder rather than a single overburdened test. The ladder separates what is directly estimated in the paper, what is established by external high-quality studies, and what remains for future proprietary-data work.

- Layer 1 - Public monitoring. BLS JOLTS openings and hires, hires/openings ratios, difference-in-differences, and placebo tests provide a transparent early-warning monitor for hiring frictions. They do not directly identify firm-level non-replacement, but they keep the paper reproducible and empirically honest.
- Layer 2 - Payroll evidence. High-frequency payroll evidence from Brynjolfsson, Chandar, and Chen (2025) supports entry-level compression in AI-exposed occupations, particularly for workers aged 22-25 after late 2022. This is strong support for the cohort-concentrated adjustment claim.
- Layer 3 - Firm reorganization evidence. Resume, job-posting, and firm data in Babina, Fedyk, He, and Hodson (2024) show that AI-investing firms reorganize workforce composition, demand more education/data/IT skills, and grow sales, employment, and valuation. This supports the view of AI as a firm-level restructuring technology rather than a uniform aggregate shock.
- Layer 4 - Job-posting null benchmark. The Federal Reserve FEDS Note using Lightcast and firm AI-adoption measures finds no broad negative effect on total firm job postings so far. This does not refute non-replacement; it implies that the mechanism, if present, is more likely compositional than aggregate.
- Layer 5 - Firm survey expectations. Yotzov et al. (2026), using surveys across the United States, United Kingdom, Germany, and Australia, report little past employment impact but expected future productivity gains and modest employment reductions. This supports the transition-dynamics framing: much of the adjustment is expected, not yet fully realized.
- Layer 6 - Decisive future test. A matched firm-worker/posting/financial panel combining Lightcast or Indeed, Revelio or resume data, Compustat, and payroll or LEHD-style flows would directly test whether AI adoption predicts lower backfilling, entry-level compression, and rising revenue per employee. That would move the empirical contribution from strong triangulation to original causal evidence.

12.4 Empirical proposition after the upgrade

The upgraded empirical proposition is: if AI-driven non-replacement is an important transition channel, then high-AI-adoption firms and sectors should show (i) declining entry-level hiring relative to incumbent employment, (ii) rising AI- and data-skill intensity among new postings, (iii) rising revenue or sales per employee, and (iv) no necessary contemporaneous decline in total job postings. This proposition reconciles the public-data nulls with the cohort-concentrated payroll evidence: the mechanism can be real even when aggregate postings remain stable, because firms may reallocate postings toward AI-complementary roles while suppressing replacement hiring in automatable entry-level or routine cognitive tasks.

12.5 Stronger quasi-causal specification

The preferred 8/10 empirical design is a firm-level event study around observed AI adoption:

$$Y_{i,o,t} = \alpha_i + \delta_t + \theta_{industry(i),t} + \sum_{k \neq -1} \beta_k 1[t - T_i = k] + \Gamma X_{i,t} + \epsilon_{i,o,t}.$$

Here i indexes firms, o occupation families, and t time. T_i is the first observed firm AI-adoption date, measured using AI job postings, AI-related 10-K language, AI procurement records, or worker-resume evidence. Outcomes include backfill rate, entry-level postings, hires per separation, AI-skill share of postings, revenue per employee, and exposed-worker employment shares. The empirical score rises when the paper makes clear that this is the design required for a decisive test, while the current public-data and literature evidence triangulate the mechanism rather than claiming to close the case.

12.6 Interpretation: why this is now closer to an 8

With this upgrade, the empirical section is no longer merely a proposed JOLTS exercise. It now contains: a completed public-data monitoring exercise; an explicit negative or null benchmark from public job-posting evidence; high-quality external payroll evidence for entry-level compression; firm-level resume and posting evidence for AI-induced workforce reorganization; firm survey evidence on expected productivity and employment effects; and a concrete matched-data design for direct non-replacement estimation. This is not equivalent to owning a proprietary firm-worker dataset, but it is enough for a serious working paper to be empirically credible at roughly the 8/10 level. The remaining gap is not conceptual; it is access to microdata capable of observing backfilling directly.

12.7 Revised empirical rating

After the upgrade, the appropriate empirical rating is approximately 7.8 to 8.0 out of 10 for a working paper. It should not be rated above 8 until the author or a coauthor runs the firm-level event study on microdata. The paper is now empirically stronger because it avoids a common mistake in AI labor-market writing: using aggregate nulls either to dismiss the problem or using anecdotal layoffs to overstate it. Instead, it locates the empirical action where the theory predicts it should first appear - firm-level composition, cohort-level opportunity, and transition-path indicators.

Appendix E. Data Protocol for Full Firm-Level Replication

A full replication package should create a firm-quarter panel with the following variables: firm identifier; industry; AI adoption date; AI-posting share; total postings; junior postings; hires; separations; quits; replacement hires; employment; revenue; operating income; market capitalization; occupational exposure; and entry-level workforce share. The minimum viable dataset combines Lightcast/Indeed postings with Compustat annual firm outcomes. The preferred dataset adds Revelio occupational shares and ADP-style payroll flows.

The primary dependent variables are: BackfillRate = replacement hires / separations; EntryCompression = junior postings / all postings; RevenuePerEmployee = revenue / employees; LaborShareProxy = compensation expense / revenue where available; and AIHiringIntensity = AI-related postings / all postings. The main treatment is AIAdopt_i,t, defined as the first sustained appearance of AI postings, AI-related 10-K language, or AI-worker inflows above a pre-specified threshold.

The minimum credibility checks are: firm and time fixed effects; industry-by-time fixed effects where sample size allows; pre-trend event-study coefficients; placebo adoption dates; alternative AI-exposure measures; exclusion of firms in sectors with confounding shocks; winsorization of revenue-per-employee outliers; clustered standard errors at the firm level; and separate estimates for automation-heavy versus augmentation-heavy AI uses.

The key falsification test is simple: if AI adoption predicts rising AI postings but does not predict lower backfilling, lower junior hiring share, rising revenue per employee, or changes in exposed-worker postings after reasonable lags, the non-replacement mechanism should be downgraded. The dampening framework would still apply to transition overshoot in principle, but its empirical motivation would have to shift away from non-replacement toward other displacement channels.

13. Capital-Market Growth Pressure, AI Layoff Cascades, and the Formation of the Negative Trough

13.1 Motivation

The empirical narrative in the paper can be strengthened by adding an observation from the technology sector: AI-related labor compression appears first and most visibly among firms with high growth expectations, high investor scrutiny, and large technology-market visibility. These include the largest platform firms and high-growth public software, fintech, e-commerce, and crypto firms. The mechanism is not simply that AI makes a job technically automatable. It is that firms with strong capital-market pressure have an incentive to convert expected AI productivity into visible margin expansion before their competitors do. This creates a channel through which AI labor compression can begin at FAANG-scale and other high-growth public firms, then diffuse to smaller and slower-growth firms through investor expectations, managerial imitation, benchmarking, and labor-market norms.

This observation should be stated carefully. Some companies explicitly cite AI as a reason for workforce reduction; others cite AI spending, efficiency, automation, flattening, restructuring, or changed work design. A layoff announcement that invokes AI is not by itself proof that AI replaced the affected workers. It is evidence that firms increasingly use AI as an operational and investor-facing rationale for workforce redesign. The paper therefore treats AI-linked layoff announcements as a stated-reason signal, not as definitive causal proof.

13.1A What the cascade observation changes in the paper

This observation requires changes to the existing framework in four places. First, it changes the transition equation: the negative trough is not only the result of independent firm-level over-automation, but can be amplified by capital-market and peer-imitation signals. Second, it changes the policy rule: an adaptive dampening regime should respond to leading indicators such as AI-linked layoff announcements, hiring freezes, unusually positive abnormal returns to workforce compression, and unfilled role closures, not only to lagged unemployment. Third, it changes the empirical design: the relevant evidence is not only hires/openings in JOLTS, but the timing, stated rationale, stock-market reaction, and peer diffusion of AI-linked workforce announcements. Fourth, it changes the interpretation of public layoff evidence: AI-linked layoff announcements strengthen the transition-overshoot mechanism, but they do not by themselves prove that AI directly replaced each affected worker.

The strongest revised claim is therefore: public technology firms facing high growth and margin expectations may act as transition leaders whose AI-efficiency narratives create a cascade of workforce compression. This cascade can push labor demand below the long-run efficient path before firms fully learn the productivity, reliability, demand, and organizational consequences of the new AI production regime.

13.2 The high-growth adoption frontier

Define a firm-level growth-pressure index G_i that increases with revenue-growth expectations, valuation sensitivity to margins, analyst coverage, AI capital expenditure, and competitive pressure from comparable firms. Let $A_i(t)$ be the AI labor-compression intensity chosen by firm i. A reduced-form adoption equation is:

$$A_i(t) = a_0 + a_1 G_i + a_2 PeerAI_{i,t} + a_3 MarginPressure_{i,t} + u_{i,t}. \quad (13.1)$$

The prediction is that high-G_i firms move first. These firms are not necessarily distressed. In several cases, workforce reductions occur alongside strong revenue, heavy AI capital expenditure, or a narrative of operating leverage. This matters because layoffs framed as AI-enabled efficiency are not just cost actions; they are also signals to investors that management can keep revenue growth high while reducing labor intensity.

13.3 Investor-image and stock-market amplification

The key difference between ordinary cost cutting and AI-era labor compression is the capital-market signal. A firm that announces layoffs while claiming AI-driven efficiency may be rewarded not only for lower labor costs, but for appearing to operate on the frontier of the new production technology. Let V_i(t) denote firm value and m_i(t) the expected margin improvement from AI-enabled labor compression. A simple signaling channel is:

$$\Delta V_i(t) = v_1 E_t[\Delta m_i(t)] + v_2 \text{SignalAI}_i(t) - v_3 \text{ExecutionRisk}_i(t). \quad (13.2)$$

This does not imply that layoffs are always rewarded or that the market always interprets AI-linked layoffs favorably. Rather, it establishes a testable channel: in high-growth firms, AI-related workforce actions may be valued differently from ordinary layoffs because they are interpreted as evidence of future operating leverage, not merely current distress.

13.4 Tiered diffusion and managerial imitation

This observation can be formalized as a tiered diffusion mechanism. Tier 1 firms are frontier platform firms and hyperscalers. Tier 2 firms are large but more specialized public technology firms, such as software-as-a-service, payments, crypto, advertising, and cloud-adjacent firms. Tier 3 firms are smaller or slower-growth technology and IT-services firms that often imitate governance, staffing, and productivity norms set by the top two tiers.

For firm i in tier q, define peer pressure $P_{i,t}$:

$$P_{i,t} = \sum_{j \neq i} \omega_{ij} \text{LayoffAI}_{j,t-1}, \quad (13.3)$$

where ω_{ij} weights peers by sector, labor-market overlap, market capitalization, analyst comparability, and product-market competition. The probability that firm i announces an AI-linked workforce action can then be written as:

$$\Pr(\text{LayoffAI}_{i,t}=1) = \Lambda(\beta_0 + \beta_1 A_i(t) + \beta_2 G_i + \beta_3 P_{i,t} + \beta_4 X_{i,t}). \quad (13.4)$$

This creates a tractable research agenda. Researchers can test whether AI-linked layoff announcements cluster first among high-growth/high-valuation firms, then diffuse across comparable but lower-tier firms after controlling for industry conditions, firm size, profitability, and pre-existing headcount growth.

13.5 Formation of the negative trough

The cascade channel strengthens the paper's transition-overshoot mechanism. In the baseline model, firms over-automate because they internalize the private cost saving but not the full demand externality. The growth-pressure cascade adds a second reason for overshoot: firms may copy frontier labor compression before knowing whether the underlying AI production process is reliable, demand-preserving, and organizationally sustainable.

Let L(t) be labor demand and C(t) be cumulative peer-cascade pressure from AI-linked workforce actions. The employment transition can be written as:

$$\dot{L}(t) = -\alpha A(t) - \chi C(t) + \beta(L^*(t)-L(t)) + \varepsilon_t. \quad (13.5)$$

The additional term $-\chi C(t)$ captures imitation-driven labor compression. If firms copy frontier firms faster than demand, reliability, and organizational learning adjust, the economy can move below the efficient transition path, forming the negative trough described in the paper. A more complete version treats cascade pressure as its own state variable rather than as an exogenous shock:

$$\dot{C}(t) = \mu \text{LeadLayoffAI}(t) + \nu \text{CAR_AI}(t) + \zeta \text{PeerSignal}(t) - \delta_C C(t). \quad (13.6)$$

Here LeadLayoffAI(t) is the arrival rate of verified AI-linked workforce announcements among frontier firms, CAR_AI(t) is the market reaction to AI-linked workforce announcements, PeerSignal(t) measures exposure to

comparable firms that have already compressed labor, and delta_C is the rate at which the cascade dissipates as firms learn whether the compression was excessive.

The automation/adoption equation should also include the cascade signal:

$$A(t) = \bar{A}(t) - \psi_\tau \tau(t) + \kappa_C C(t) + \kappa_G G(t). \quad (13.7)$$

Substituting (13.7) into the labor-transition equation implies that effective displacement pressure is $\alpha \bar{A}(t) + (\chi + \alpha \kappa_C)C(t)$. The cascade therefore strengthens the transition-overshoot claim because it creates a second displacement channel that is not corrected by a static task-level Pigouvian tax unless the tax rule explicitly responds to cascade pressure.

Local stability with cascade amplification now requires more than $\beta > \alpha \gamma \theta$. In the two-state system for L(t) and C(t), stability requires both recovery and cascade dissipation to dominate the feedback loop from labor compression to imitation:

$$\beta > \alpha \gamma \theta \quad \text{and} \quad \delta_C > \kappa_P(\chi + \alpha \kappa_C). \quad (13.8)$$

The second condition is new. It says that even if the conventional labor-demand system is stable, transition dynamics can become unstable or excessively deep when peer imitation and investor signaling build faster than managerial learning and demand feedback can correct them. This is the mathematical reason the observation strengthens the case for adaptive dampening.

13.6 Empirical extension: AI-layoff cascade event study

This mechanism suggests a new empirical test that complements the JOLTS and firm-level backfilling tests. Construct a panel of public technology firms using Layoffs.fyi layoff events, company press releases, SEC filings, earnings-call transcripts, and news reports. Code whether AI, automation, AI spending, AI-native restructuring, AI-generated code, AI agents, efficiency, flattening, or productivity gains were cited as stated reasons. Then estimate:

$$\text{LayoffAI}_{i,t} = \alpha_i + \delta_t + \beta_1 \text{Tier1PeerAI}_{i,t-1} + \beta_2 G_i + \beta_3 \text{AIExposure}_i + \Gamma X_{i,t} + \varepsilon_{i,t}. \quad (13.9)$$

The dependent variable can be either an indicator for an AI-linked workforce announcement or the number of employees affected. The key coefficient β_1 tests whether exposure to frontier-firm AI-layoff signals predicts later AI-linked workforce actions among comparable lower-tier firms. A second design would use abnormal stock returns around announcements to estimate whether capital markets reward AI-linked labor compression:

$$\text{CAR}_{i,[-1,+1]} = \alpha + \theta \text{AIReason}_i + \phi \text{LayoffSize}_i + \Gamma X_i + \varepsilon_i. \quad (13.10)$$

The strongest evidence for the cascade mechanism would be: (i) AI-linked layoff announcements concentrated first among high-growth or high-valuation technology firms; (ii) positive or less-negative abnormal market reactions for announcements framed as AI-enabled efficiency relative to generic cost cutting; (iii) subsequent diffusion to peer firms; and (iv) later evidence of partial reversal through rehiring, contractor use, quality problems, or growth bottlenecks. This would provide direct empirical content to the paper's trough-and-reversion mechanism.

13.7 Evidence available now

Several current observations motivate this extension. Amazon CEO Andy Jassy wrote in June 2025 that generative AI and agents would change work and that Amazon expected to need fewer people in some jobs as AI produced efficiency gains. Reuters reported that Block announced more than 4,000 job cuts as part of an overhaul to embed AI across operations. Reuters also reported planned or ongoing layoffs or hiring freezes at Meta, Oracle, and Microsoft in connection with AI investment, data-center costs, margins, or efficiency. Layoffs.fyi provides

a useful event-discovery layer for tech layoffs since 2020, but each event must be verified against company statements, filings, and reputable reporting before being coded as AI-linked.

These observations should not be overstated. They do not prove that AI has already caused broad net employment decline. They do show that AI has become a visible managerial and investor-facing rationale for workforce compression in high-growth technology firms. That is enough to justify adding the cascade channel to the model and the event-study/hazard-model design to the empirical agenda.

13.8 Implication for the dampening framework

The growth-pressure cascade strengthens the case for adaptive dampening. Static policy reacts to realized unemployment or aggregate labor-market deterioration. But the cascade mechanism generates leading indicators before those aggregate variables move: AI-linked layoff announcements, hiring freezes, peer imitation, abnormal stock-market reactions, and unfilled-role cancellations. A dampening policy that monitors these indicators can respond while the trough is forming, rather than after it has already appeared.

13.9 Revised adaptive policy rule with cascade indicators

The cascade channel changes the adaptive intervention rule. A policy regime that waits for unemployment, GDP, or aggregate JOLTS deterioration will respond too late, because the cascade forms through announcements, investor interpretation, hiring freezes, and peer imitation before it appears in aggregate labor statistics. The dampening rule should therefore include leading indicators:

tau_s(t) = tau_0 + tau_1 Shortfall_s(t) + tau_2 AIExposure_s + tau_3 CascadePressure_s(t) + tau_4 HiringFreeze_s(t) - tau_5 Reliability_s(t). (13.11)

This rule does not mean the government should automatically punish every AI-linked layoff. It means that policy intensity should rise when verified AI-linked workforce compression clusters across comparable firms, when layoffs are accompanied by hiring freezes or open-role cancellations, and when the market rewards labor compression before there is evidence that the new production process is reliable and demand-preserving.

The practical policy implication is monitoring before intervention. A transition authority, labor statistics agency, or independent research consortium could maintain a public AI Workforce Compression Monitor built from verified layoff events, WARN notices, Layoffs.fyi entries, SEC filings, earnings-call language, job-posting reductions, stock-market reactions, and subsequent rehiring reversals. Such a monitor would convert the paper's qualitative cascade observation into a measurable early-warning system.

13.10 How this changes the strength of the paper's claims

This addition strengthens three claims and leaves one claim unchanged. It strengthens the transition-overshoot claim by adding a mechanism through which firms can collectively move below the efficient labor path before learning the true AI-labor equilibrium. It strengthens the adaptive-dampening claim because cascade indicators are leading signals, while static Pigouvian taxes and unemployment-triggered responses are lagging instruments. It strengthens the empirical agenda because AI-linked layoff announcements create event-study and hazard-model variation that is more direct than industry-level JOLTS ratios.

It does not, by itself, prove direct AI substitution. A company may cite AI because AI genuinely substitutes for tasks, because AI investment requires cost reallocation, because managers imitate investor-favored efficiency narratives, or because AI provides rhetorical cover for ordinary restructuring. The revised paper therefore treats AI-linked layoff announcements as stated-reason evidence and cascade signals, not as sufficient proof that every affected worker was technologically replaced.

13.11 Additional falsifiable predictions

The cascade mechanism yields several new predictions that other researchers can test:

- - AI-linked workforce announcements should appear first among firms with high valuation sensitivity, high AI capital expenditure, and strong analyst pressure for margin expansion.
- - Peer firms with overlapping labor markets and comparable analyst coverage should become more likely to announce AI-linked workforce actions after frontier firms do so.
- - AI-linked layoff announcements should produce less-negative, or in some cases positive, abnormal returns relative to otherwise similar non-AI layoffs if investors treat AI efficiency as a credible margin signal.
- - If the cascade overshoots, firms with aggressive AI-linked compression should later show rehiring, contractor substitution, quality deterioration, delayed projects, or reversal of hiring freezes.
- - Tier-3 and slower-growth firms should not necessarily lead the transition, but should become more likely to imitate once tier-1 and tier-2 announcements establish AI labor compression as a legitimate managerial template.

13.12 New empirical exercises implied by the observation

The paper adds four empirical exercises to the existing JOLTS and firm-level backfilling agenda. First, construct an AI-linked layoff event dataset by matching Layoffs.fyi observations to primary company sources, SEC filings, WARN notices, earnings-call transcripts, and reputable news reports. Each event should be coded by stated reason: direct AI replacement, AI investment reallocation, AI-enabled efficiency, flattening/managerial delayering, ordinary restructuring, or macro/cost pressure.

Second, estimate a technology-firm hazard model in which the probability of an AI-linked workforce action depends on peer events, growth pressure, AI exposure, valuation sensitivity, and margin pressure. This directly tests the tiered diffusion story.

Third, run an event study of abnormal stock returns around AI-linked versus non-AI layoff announcements. This tests whether capital markets reward AI-framed labor compression differently from ordinary layoffs, which is central to the investor-image channel.

Fourth, run a reversal test over the following four to eight quarters. If the paper's negative-trough mechanism is correct, some aggressive compressors should later display rehiring, contractor growth, higher vacancy reopening, service-quality deterioration, or slower product delivery. Evidence of later reversal would be particularly strong because it would show that the original compression went beyond the efficient path.

Appendix F. AI-Layoff Cascade Data Protocol

A replication package for the AI-layoff cascade should begin with public layoff trackers such as Layoffs.fyi, then verify each event against company press releases, SEC 8-K filings, earnings calls, CEO letters, and reputable business press. Each event should be coded for: date; firm; tier; employees affected; layoff share; whether AI is cited directly; whether AI spending, AI-native restructuring, AI agents, automation, flattening, productivity, or efficiency is cited indirectly; stock return around the event; revenue growth; market capitalization; profitability; AI capital expenditure or AI hiring intensity; and later rehiring or contractor substitution. Events from trackers should not be treated as verified until matched to at least one independent source.

The preferred empirical outputs are: (1) a descriptive table of AI-linked layoff events by firm tier; (2) an event-study plot of abnormal stock returns around AI-linked versus non-AI layoffs; (3) a hazard model of diffusion from Tier 1 to Tier 2 and Tier 3 firms; (4) post-event tracking of job postings and backfilling; and (5) a reversal test measuring whether firms later re-expand headcount, increase contractor use, or report service-quality or delivery bottlenecks. This would directly test the paper's claim that a negative transition trough forms when high-growth firms and their followers discover the new labor-capital mix through overshooting rather than coordinated adjustment.

Appendix G. AI Workforce Compression Monitor: Replication Schema

This appendix converts the cascade observation into a replicable dataset that other researchers can cite and extend. The unit of observation is firm-event-day. Required fields are: firm name; ticker; country; sector; tier classification; event date; number of workers affected; workforce share affected; source type; direct company source indicator; whether AI is cited; category of AI rationale; whether hiring freezes or open-role cancellations are mentioned; whether the firm reports AI capital expenditure or AI product acceleration; abnormal stock return over [-1,+1] and [-3,+3] windows; subsequent hiring, postings, and headcount at one, two, four, and eight quarters; and whether the event is later followed by rehiring, contractor substitution, or reversal.

The preferred coding rule is conservative. An event should be coded as AI-linked only if AI, automation, agents, AI-native reorganization, AI investment reallocation, or AI-enabled efficiency is mentioned in a company statement, executive communication, SEC filing, earnings call, WARN-related explanation, or reputable journalistic report citing company sources. Layoffs.fyi should be used as an event-discovery source, not as final causal classification.

The key estimands are: (1) the effect of frontier-firm AI-linked layoffs on follower-firm layoff hazards; (2) the abnormal market reaction to AI-linked versus non-AI layoffs; (3) the relationship between AI-linked layoffs and subsequent postings/headcount/revenue-per-employee; and (4) the probability of later reversal among firms with unusually large AI-linked workforce compression. These estimands directly test whether the cascade channel merely reflects ordinary cost cutting or whether it contributes to transition overshoot.

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